Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for

Analgesia and Anaesthesia in Maternity

Northern Services

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1. Purpose

1.1. The purpose of this document is to detail the process for providing analgesia for women in labour and anaesthesia in the peripartum period in accordance with best practice guidance.

1.2. The guideline applies to midwives, obstetricians and anaesthetists. Non compliance with this guideline may be for valid clinical reasons only. The reason for non compliance must be documented clearly in the medical notes.

1.3. Implementation of this guideline will ensure that clinicians are aware of the minimum standard of practice set out within this document.

2. Abbreviations

TENS	Transcutaneous Electrical Nerve Stimulation
PCA	Patient Controlled Analgesia
MEOWS chart	Modified Early Obstetric Warning System chart
CTG	Cardiotocograph
ABC	Airway Breathing Circulation
PCEA	Patient controlled Epidural Analgesia
BMI	Body Mass Index
LMWH	Low Molecular Weight Heparin

3. Responsibilities

Role of Clinicians

3.1. The Midwife is responsible for:

- Being an advocate for women under their care and promoting individual choice
- Ensuring that pain relief is administered promptly
- Referring to the Obstetrician and Obstetric Anaesthetist (as appropriate) to request assessment of pain relief options/anaesthesia
- Recording fluid input and output

- Ensuring ongoing monitoring and observations are recorded, scored and escalated as appropriate

3.2. The Obstetrician is responsible for:

- Ensuring that women are fully informed of their pain relief options
- Ensuring that women have appropriate and adequate analgesia prescribed, fully and accurately
- Undertaking liaison with the Midwifery team and Obstetric Anaesthetist for assessment of anaesthesia/pain relief options
- Comprehensively recording a plan for analgesia in the notes
- Ensuring that on-going assessment, planning and treatment are undertaken frequently and documented clearly

3.3. The Obstetric Anaesthetist is responsible for:

- Ensuring that women are fully informed of their pain relief options
- Ensuring that women have appropriate and adequate analgesia prescribed, fully and accurately
- Undertaking liaison with the Midwifery team and Obstetrician for assessment of anaesthesia/pain relief options
- Comprehensively recording a plan for analgesia in the notes
- Ensuring that on-going assessment, planning and treatment are undertaken frequently and documented clearly

Role of Maternity Services Governance Group

3.4. The Maternity Services Governance Group is responsible for:

- Assessing need for updates to guidelines
- Reviewing updated versions of guidelines/any new guidelines
- Approving updated/new guidelines

4. Analgesia for labouring women - non regional

4.1. Breathing exercises, relaxation techniques, immersion in water, hypnobirthing and massage may reduce pain in the latent stage of labour: women should be supported in their choice. If a woman chooses to use massage techniques in labour that have been taught to birth companions, she should be supported in this choice.

4.2. Do not offer or advise the use of hypnobirthing, aromatherapy, yoga or acupressure for pain relief in the latent stage unless you have been appropriately trained in these techniques.

4.3. Support the playing of music of the woman's choice in labour.

Transcutaneous Electrical Nerve Stimulation - TENS

4.4. This form of pain relief is based upon two hypotheses of pain physiology: the 'gate control theory' and the production of stimulation of endogenous opiates.

4.5. A TENS unit is an electronic stimulus generator which transmits pulses of various configurations to electrodes which are attached to the skin on the woman's back, overlying the thoracic(T10) and lumbar (L1) nerve endings and over the sacral nerves (S2-4). Accurate placing of the electrodes is important for maximising the pain relief.

4.6. The woman operates the equipment herself after instruction on how to increase stimulation strength and frequency levels and how to use the additional button for continuous stimulation for use during contractions and how to reset to pulse again afterwards. It is most effective when commenced in early labour.

4.7. NICE do not recommend offering TENS to women already in established labour.

4.8. There are no side effects and TENS can be used in conjunction with some other forms of pharmacological pain relief (NOT with water).

Labour and Birth in Water

4.9. Labouring in water is supported for healthy women with uncomplicated pregnancies at term and is recommended for pain relief. Immersion in water during the first stage of labour significantly reduces women's perception of pain and use of epidural analgesia without adversely affecting labour duration, operative delivery rates or neonatal wellbeing.

4.10. Indications for using the pool (see also Intrapartum care: care of healthy women and their babies during childbirth Including Fetal Monitoring in Labour)

4.10.1. The following criteria should be used as a guide to assess the suitability of a woman who chooses to use the birth pool:

- Woman's informed choice
- Term pregnancy >/= 37 weeks gestation
- Cephalic singleton pregnancy
- Clear liquor
- Normal maternal and foetal observations
- At least two hours since administration of opioids
- No known or suspected infection

- Established labour – however the pool may be useful in latent phase of labour for relaxation and pain relief: the woman should be encouraged to leave the pool to become established in labour
- Spontaneous onset of labour. Women who are undergoing an induction of labour for post-dates with no other risk factors may use the pool, providing they meet the above criteria, have had a normal CTG and do not require further intervention such as oxytocin to establish labour.

4.11. Telemetry: Management of high risk women in labour and birth using Water Birthing Pool and Telemetry Standard Operating Procedure

- Women who request labour and/or delivery in the pool who do not meet the stated criteria must have a clearly documented discussion and plan with a senior Obstetrician and Midwife on an individualised basis to determine the risk. Detailed notes should be made of any discussions. See "Telemetry: management of high risk women in labour and birth using water birthing pool and telemetry standard operating procedure"

4.12. Prior to entering the pool

- Women should be given information to enable them to make an informed choice, with documentation in the notes. Discuss the reasons why the woman may be asked to leave the pool

- All equipment should be checked and the completed check list sticker placed in the intrapartum notes.

Hoist Check	
Woman Identity	
Woman's weight	
Midwife	
Consent	
Correct sling	
Up/down mechanism	
Battery	
Inspection label	
Equipment damage/fraying	
Date	
signature	

4.13. Observations

- Ambient room temperature should be comfortable
- Encourage drinking to avoid dehydration
- Times of entering and leaving the pool should be documented in the intrapartum notes, including the reason for leaving the pool

- Monitor temperature of woman and of the water hourly. The temperature of the woman should not be above 37.5 degrees C.

4.14. Management of the second stage of labour

- Never leave the woman in the second stage of labour.
- Ensure her bottom is under the water for delivery. If she raises herself out of the water once the foetal head is out, she should remain out of the water to complete the birth. Shoulders would be expected to be delivered within one contraction following the birth of the foetal head. Gentle downward traction may be used to facilitate the delivery of the shoulders if needed. If delivery is not completed within one contraction after delivery of the foetal head, rapidly initiate active management.
- The midwife should conduct the delivery using a 'hands off' approach to ensure no stimulation to gasp is caused, bringing the baby to the surface with the face upper-most.
- Keep the baby's body immersed in the water to ensure the baby's temperature is maintained.
- If the cord is around the baby's neck and/or body, disentangle under the water
- The midwife should be prepared for the possibility that the cord may snap. If this happens both ends must be clamped urgently.
- Any deviation from the physiology of normal labour should alert the midwife to consider if it is still safe for the woman to remain in the pool
- Any signs of foetal distress should be communicated to the woman and she should be asked to leave the pool for appropriate monitoring of foetal wellbeing.
- An episiotomy should not be performed in the water
- In the event of shoulder dystocia, encourage the woman to leave the pool. The change of position may be sufficient to dis-impact the shoulders. If necessary transfer to the bed and follow the Shoulder Dystocia Guidelines.

4.15. Management of the Third Stage of Labour

- May follow the guidelines for physiological third stage after having fully discussed differences and recommendations between active management and physiological management of third stage. Active management should be undertaken out of the pool
- If the amount of bleeding seems excessive the woman should be asked to leave the pool and the appropriate management undertaken to minimise blood loss
- Never cut the umbilical cord under the water

- If the woman's condition permits, perineal suturing should be delayed for up to one hour to allow the tissue to revitalise after water immersion

4.16. Cleaning

- Keep birthing pools clean using protocol agreed with the microbiology department and in accordance with manufacturer's guidelines. Infection control state that the pool should be rinsed and debris removed, followed by cleaning with Tristel Solution, then cleaning with a new mop head which is disposed of afterwards. If the pools are not in use, they are to be flushed daily with the taps left running for three minutes.

Entonox

4.17. Entonox is a mixture of 50% oxygen and 50% nitrous oxide, and is available in all birth settings for pain relief in labour

4.18. Women should be informed that it may decrease pain in labour but may make her feel nauseous and light-headed

4.19. Has no known side effects for the foetus

4.20. Entonox analgesic effect begins in around 20 seconds, with maximum effect after 45-50 seconds

4.21. Woman should take controlled deep breaths of entonox as soon as she feels the contraction starting and continue until it starts to subside. She should then be encouraged to take fresh air and sips of water

4.22. A single use filtered mouthpiece and new scavenger tubing must be used for each woman and discarded after the birth

4.23. May be used in combination with other pain relief eg water, TENS

Intramuscular (im) opioids

4.24. Diamorphine is the intramuscular opioid of choice, and is available in all birth settings

4.25. Provide limited pain relief in labour

4.26. May have significant side effects for both woman and foetus

- Respiratory depression
- Drowsiness
- Nausea and vomiting
- Urinary retention
- Itch
- Respiratory depression and drowsiness in the baby may last several days
- May interfere with breast feeding

4.27. Antiemetic is usually administered with im opioid eg cyclizine 50 mg in 1 ml by im injection

4.28. Women should not enter the birthing pool within 2 hours of an injection or if they feel drowsy

4.29. Vaginal examination is recommended before giving im opioid, to confirm progress and stage of labour. There is potential for maximum effect of the opioid to coincide with delivery, and side effects for the baby

4.30. Up to two doses of diamorphine can be given as a 'midwives exemption' in established labour. Dose is 5-10 mg intramuscular injection, no less than 4 hours apart. Further or larger doses are rarely used, and may only be given on doctor's prescription and after discussion with a Senior Obstetrician and usually also with the Duty Obstetric Anaesthetist.

4.31. When considering use of opioids, the woman's medical history, and any regular opioid use/ possible tolerance should be considered. Contraindications should be checked for and documented. If in doubt, discuss with Senior Obstetrician and/or Obstetric Anaesthetist.

Remifentanil iv Patient Controlled Analgesia PCA

4.32. General Principles

- Remifentanil is a short acting synthetic opioid analgesic. It provides effective analgesia during labour. It is administered intravenously via a PCA pump. Consent, prescription and set up is by the Obstetric Anaesthetist. It has rapid onset and short duration of action and no active metabolites.

- Like all opioids, can cause severe respiratory depression and sedation. Careful explanation to the woman and appropriate monitoring is essential. For this reason, remifentanil is generally only offered to those patients in whom epidural analgesia is contraindicated or has failed.

- The General On Call Consultant Anaesthetist and the Duty Obstetrician must be informed if any woman is using remifentanil PCA.

4.33. Indications

- Alternative to im opioids for pain relief in labour when epidural analgesia is contraindicated or has failed

4.34. Contraindications

- Allergy to remifentanil
- Multiple pregnancy
- Other parenteral opioid administration in the last 2 hours
- Less than 36 completed weeks' gestation (ie only for 37 + weeks gestation). May be used before this gestation only if consultant obstetrician has documented non-viable foetus or another specific reason for remifentanil use.
- Trained midwife must be available to provide 1:1 care throughout

4.35. Preparation of the woman

should read the remifentanil PCA information leaflet

- should be informed of possible side effects as set out in the leaflet, including drowsiness, itch, nausea and dizziness
- must be informed that approx. 1:10 women using it will have lowered oxygen saturation levels and may require additional oxygen
- must be informed that has been associated with significant / concerning respiratory depression and sedation
- A dedicated iv cannula is required: NO other medication or fluid should be administered through this cannula at any time
- Must have separate iv cannula which is used in the normal way. The cannulae may be in the same arm.
- Should be shown how to use the PCA, and told to press the PCA button just before or at the start of a contraction
- Must have pulse oximeter monitoring in place before PCA is started and used continuously throughout.

- Oxygen must be available in the delivery room, along with Hudson type mask and nasal cannulae. A resuscitation type Bag Valve Mask (BVM) and oxygen tubing must be IN the delivery room, as well as all other resuscitation equipment readily available in the Delivery Suite.

4.36. Equipment

- 50 ml bag 0.9% saline
- 2 mg remifentanil from Controlled Drugs Cupboard in Theatre 8
- Dedicated Remifentanil PCA pump (BBraun type, blue front): pre-programmed to deliver 20 mcg (0.5 ml), 30 mcg (0.75 ml) or 40 mcg (1.0 ml) bolus over 6 seconds with a 2 minute lockout. PCA handset and locking key. Pumps etc are kept in the equipment store in Delivery Suite.
- 50 ml luer lock syringe
- iv PCA extension set with non-siphon valve
- Pulse oximeter and other routine monitoring
- Resuscitation equipment available on the 'cardiac arrest' trolley for the Delivery Suite. A BagValveMask BVM and oxygen tubing must be IN the delivery room.
- Separate iv line with iv fluid prescribed
- Remifentanil iv PCA monitoring pack: MEOWS chart, remifentanil PCA flowchart, patient information leaflet, audit sheet. See Appendix 2.

4.37. Setting up

4.37..1. Syringe Preparation by Obstetric Anaesthetist

Reconstitute 2 mg remifentanil with 50 ml 0.9% Saline: 40 mcg per ml solution, which is stable for 24 hours at room temperature

4.37..2. Programming and Setting up the PCA pump

- Obstetric Anaesthetist will prescribe a bolus dose of 20, 30 or 40 mcg remifentanil via iv PCA with a lockout time of 2 minutes
- 20 mcg bolus should be used initially by all women.

- Effective dose will depend on stage of labour and on eg size of woman: dose may need to be adjusted by the anaesthetist during labour. The maximum recommended dose is 0.5 mcg per kg lean weight.

- Anaesthetist will prescribe oxygen as required for target SpO2 96-100%

- PCA pump syringe clamp must be locked in place when the pump is in use to prevent any tampering with the syringe (see underside of pump). A key is held by the midwifery team. The pump keypad must also be locked by the anaesthetist using the allocated code.

- The anaesthetist must stay in the Delivery Suite for at least the first 20 minutes of remifentanil PCA use, and for at least 20 minutes after any dose change.

- The woman will need some help and supervision specifically to get the timing right so that the remifentanil gives pain relief for the contraction and to avoid sedation and respiratory depression

4.38. Managing the PCA pump

4.38..1. Changing the syringe

The syringe is changed by the anaesthetist when it is empty, making up a new syringe as above. The syringe clamp must be locked in place with the key and the pump keypad locked once the new syringe is in place. The midwife will need to contact the anaesthetist in time to allow syringe change without complete loss of pain relief due to empty syringe.

4.38..2. Changing the dose

Dose alterations can be made as labour progresses. A new dose must be prescribed by the anaesthetist, pump settings changed, prescription and pump settings checked with Midwife, and syringe clamp and keypad locks in use. The anaesthetist must remain in the Delivery Suite for at least 20 minutes after a dose change.

4.39. Observations and Monitoring

- The MEWS chart is used as the remifentanil PCA monitoring chart for woman's observations

- Use the MEWS chart together with the remifenanil PCA Flow Chart to guide monitoring and actions
- Continuous SpO2 monitoring must be established before starting PCA and continued throughout, with observations recorded on the MEOWS chart at least every 30 minutes.
- Continuous CTG monitoring may be used at the discretion of the Obstetrician and/or Midwife
- Respiratory rate must be measured just after the end of a contraction, and recorded at least every 30 minutes
- MEWS every 30 minutes

4.40. Indications for contacting the Obstetric Anaesthetist

See Flow Chart: Appendix 2

4.40..1. Remove the PCA pump handset and contact the Anaesthetist and Senior Midwife if:

- The woman is drowsy and does not rouse easily to voice
- Respiratory rate less than 10 per minute
- Sp02 remains 95% or less despite oxygen 4l/min via nasal cannula or Hudson mask

4.40..2. Contact the Obstetric Emergency Team via switchboard (Dial 2222) and consider requesting Cardiac Arrest Team (Dial 2222) if the woman rouses only to physical stimulus or to pain or is unresponsive or if Sp02 is below 90% at any time, or if the midwife is concerned about the woman. Remove the PCA handset. Begin resuscitation using the ABC approach. Consider giving naloxone 200 mcg iv.

4.40..3. Inadequate analgesia

Assess timing of pressing PCA button: needs to be pressed right at the beginning of a contraction. If analgesia inadequate despite appropriate timing, contact the Duty Obstetric Anaesthetist.

4.40..4. Any other concerns should be discussed initially with the Senior Midwife and escalated to the Anaesthetist and/or Obstetrician as indicated.

4.41. Safe Practice Points

- 1:1 care by trained midwife, at all times
- Never leave the woman alone

- Always use a dedicated cannula
- The cannula should be removed without flushing when the PCA is discontinued
- Do not give any other medication or fluid via the PCA cannula
- Only the woman may use the PCA button. It must never be pressed by the midwifery staff or birth partner.
- PCA can be used during the delivery (second and third stage) and for the repair of tears and episiotomies, but will not provide adequate pain relief, local anaesthetic must be offered as usual
- 'entonox' may be used at the same time as remifentanil PCA. If nasal oxygen is in use, this should continue alongside the Entonox.
- Naloxone must always be available in the Delivery Suite
- Due to the short duration of action of remifentanil, naloxone is very rarely required to treat overdose in this setting.
- Remifentanil shows very similar pharmacokinetics in the neonate, so naloxone is very unlikely to be needed to treat overdose in this setting
- The PCA pump keyboard should be locked (level 1 lock) when in use, so that the key symbol appears on the display. The syringe clamp must also be locked, using the key, so that the syringe cannot be removed from the pump. A key is held by the midwifery team.

Fentanyl or Morphine iv PCA for labour

4.42. Fentanyl PCA is used regularly in some centres for labour analgesia with a viable foetus. It may be considered in North Devon District Hospital after discussion with General On Call Consultant Anaesthetist and Consultant Obstetrician and Paediatrician. Specific written notes regarding consent, prescription, monitoring plan, and actions to address any complications must be recorded.

4.43. Fentanyl or morphine iv PCA may be used for labour analgesia for women who have had an intra-uterine death.

- Before commencing PCA, woman must be informed of risks and benefits, and record made of consent discussion

- She must be familiar with the concept of PCA, and know that only she may press the PCA button
- Consider adding paracetamol to reduce opioid need and side effects
- A loading dose of opioid may be needed to initiate analgesia. This may be given intravenously by Anaesthetist, and may not be required if im diamorphine has already been used.
- Caution after im opioids: consider allowing at least 2 hours after im dose before starting PCA.
- PCA should be discontinued as soon as it is possible to provide sufficient analgesia by oral route.
- Please refer to *Acute Pain Guidelines* for other details regarding morphine or fentanyl PCA.

5. Analgesia for labouring women – Regional Analgesia

Epidural Analgesia

5.1. Is provided 24 hours a day on Delivery suite by the Anaesthetic Team. It takes about 40 minutes from the arrival of the anaesthetist until effective pain relief may be established: this should be considered when discussing epidural with labouring women

5.2. Advantages and disadvantages of epidural

Advantages	Disadvantages
Most effective form of analgesia in labour	May cause some loss of sensation or motor power
Does not generally cause drowsiness or nausea (unless hypotensive)	Failure rate of 3% and imperfect in 10-15%
Minimal foetal depression	May cause loss of bladder sensation
No significant interference with uterine action	May cause hypotension
Control of hypertension	May cause itch
May be used for instrumental or operative delivery	Associated with longer second stage of labour and increased chance of instrumental delivery

Not a cause of long-term backache, though epidural stie may be tender for a few days, and may rarely last longer	Incidence of dural puncture of about 1% or which about 2/3 will get headache, which may be severe
Not associated with longer first stage of labour or increased chance of caesarean section	May cause mild pyrexia
	Rare possibility of nerve damage (1 in 1000 temporary, 1 n 13 000 permanent, 1 in 250 000 severe eg paralysed)
	Epidural abscess or haematoma or meningitis (very rare)

5.3. Indications

- Maternal request
- Failure of other methods of analgesia
- Hypertensive disorders and pre-eclampsia if platelet count/clotting normal (see below)
- Expectation of operative delivery eg twins, breech, occipitoposterior position, trial of labour (Vaginal Birth After Caesarean or VBAC), prolonged labour
- Maternal medical conditions eg respiratory disease, cardiac disease

5.4. Contraindications

Absolute Contraindications	Relative Contraindications
Maternal refusal	Spinal deformity +/- previous surgery
Platelet count less than 70 x10^9 per litre	Platelet count 70-100 x10^9 per litre (with normal clotting screen)
Local infection or severe systemic infection	Suspected/mild systemic infection/sepsis
Coagulopathy or full anticoagulation	Recent treatment with anticoagulant (see below)
Local anaesthetic allergy	Raised intracranial pressure
Hypovolaemia	Some cardiac disease
Lack of 1:1 midwifery care	

Relative contraindications should be considered in each individual case, and in discussion with a Consultant Anaesthetist.

In the unusual event that midwifery workload is such that 1:1 midwifery care cannot be provided, the epidural service will be suspended. The Delivery Suite Coordinator should inform the Duty Obstetric Anaesthetist and the General On Call Consultant Anaesthetist as well as considering escalating to Midwifery Manager on Call. No further epidurals will be sited. It will very rarely be necessary to stop the epidural of a woman which is already in place, but 1:1 care for that woman must continue. Alternative analgesia wil be offered as needed. Similarly, if the workload of the anaesthetic team is excessive, the General On Call Consultant Anaesthetist should be informed, and the epidural service may have to be suspended.

It is not always necessary to check the platelet count before siting an epidural. The anaesthetist may request a platelet count and/or coagulation screen before siting it if clinically indicated, and may prefer to wait for the result if a platelet count has already been taken as part of 'routine' bloods.

5.5. Consent

Consent must be obtained before undertaking epidural and recorded in the notes. The extent to which all risks are discussed will depend on individual circumstances but discussion should include as a minimum: failure, hypotension, motor weakness, instrumental delivery, dural puncture with severe headache, and nerve damage. If an epidural is sited in the presence of relative contraindications, specific discussion of the additional relevant risks should be recorded.

In those women who have a clear statement in their Birth Plan that they do NOT want epidural analgesia, and who are now asking for it in the course of labour, consider taking written consent for labour analgesia using a standard Trust consent form.

A few women in labour will request an epidural for pain relief and will decline any information about risks and/or benefits. An epidural may be sited at the woman's request under these circumstances, with a written record made of the woman's specific request NOT to be given information about risks.

5.6. When to initiate epidural analgesia

Labour must be definitely going to progress to delivery, but rupture of membranes need not have occurred. Usually the woman will be in 'active labour' as defined by National Institute for Clinical Excellence NICE 2007 (regular painful contractions and cervical dilatation).

An assessment of the progress of labour by Vaginal Examination should have been made. If delivery is anticipated within 60 minutes, then other methods of analgesia are usually more appropriate.

5.7. Preparation by Midwifery team

- Gather relevant information to share with the Anaesthetist eg indication, relevant medical history, previous and current analgesia, recent blood results, anaesthetic plan etc
- Contact Epidural Anaesthetist by bleeping 234.
- Give the woman the Epidural Information Card https://www.labourpains.com/Epidural-Information-Card to read
- Establish iv access, take blood tests if relevant
- Prepare crystalloid infusion of 1000 ml Plasmalyte
- Ensure the woman is wearing appropriate clothing for the procedure
- Encourage woman to empty her bladder
- Collect Epidural Trolley and a stool without wheels for the feet
- Record observations: Maternal blood pressure, foetal heart rate

5.8. Technique

- Only anaesthetist trained in the technique may site an epidural under distant supervision or unsupervised. Those anaesthetist undergoing training in the technique must be supervised appropriately
- Position the woman sitting or laterally
- Use full aseptic technique including gown, mask, hat, gloves
- Skin preparation with 0.5% chlorhexidine in alcohol solution, allow to dry
- Appropriate drapes
- Skin and deeper tissues infiltrated with local anaesthetic, taking account of maximum safe doses
- Use loss of resistance to saline, and appropriate Tuohy needle
- Leave 3-5 cm of epidural catheter in the epidural space, consider inserting further for patients with high Body Mass Index
- Check absence of cerebrospinal fluid or blood on passive flow under gravity and on aspiration

- Skin fixation, Sterile transparent dressing and tape to secure
- All connections secured and checked
- Yellow 'epidural' sticker applied to the epidural catheter

5.9. Establishing analgesia

Initial dose of local anaesthetic solution should be divided into increments, so that should the dose be delivered unintentionally either intravenously or subarachnoid, the effect will be such that woman and baby remain safe and complications are readily treatable.

Under most circumstances, use of dilute local anaesthetic solution (0.1% Levobupivacaine) with opioid produces dependable analgesia with minimal motor block, better mobility and increase maternal satisfaction compared with eg 0.25% Levobupivacaine. Use of 0.25% Levobupivacaine for labour analgesia should generally be avoided unless there are specific indications.

Initial doses may be given via the Patient Controlled Epidural Analgesia (PCEA) pump, or by hand from a syringe: the spread and analgesic effect may be greater when given by syringe, but it is important to maintain sterility of all the connections at all times. It is also critical to check before any bolus dose that this is into the epidural catheter system and NOT intravenous. Due to fatal cases of epidural drugs being administered via the wrong route, the National Patient Safety Agency has issued a patient safety alert concerning epidural injections and infusions and connectors incompatible with intravenous and other routes of administration.

http://www.nrls.npsa.nhs.uk/alerts/?entryid45=94529

However at this time the equipment is still not fully available in the UK, and care must be exercised to guard against wrong route injection. Epidural giving sets are manufactured with a yellow colour code, and yellow identification labels should be applied to any syringes intended for injection into the epidural system.

- Initial dose
  - Aspirate (for at least 30 seconds) to check for incorrectly positioned (intravascular or subarachnoid) catheter
  - Recommended initial dose 6-10 ml of 0.1% levobupivacaine with 2 mcg per ml Fentanyl

- Alternative initial dose 3 ml of 2% lidocaine (more sensitive to exclude intravascular catheter) OR 3 ml of 0.25% levobupivacaine (not generally recommended, see above)
- Blood pressure and leg mobility should be confirmed as normal 5 minutes after this initial dose

- Subsequent dose (following normal response to initial dose)
  - 10-14 ml of 0.1% Levobupivacaine with 2 mcg per ml Fentanyl in divided doses, to total loading (initial plus subsequent) dose of 20 ml
  - Alternatively 7 ml of 0.25% Levobupivacaine +/- Fentanyl up to 1 mcg per kg (lean weight) may be used under some circumstances, but is not generally recommended
  - BP and leg motor power are checked are recorded every 5 minutes for the first 20 minutes

- Setting up the PCEA pump and prescription

  The anaesthetist sets up the PCEA system, using premixed bag of 0.1% Levobupivacaine with 2 mcg per ml Fentanyl.

  There are three pre-programmed settings for use in the Maternity Unit in the epidural pump: set the Volume to Be Infused appropriately

Pump Programme Name	Background Infusion Rate	PCEA Bolus	Lockout Time
D	8 ml per hour	5 ml	15 minutes
E	5 ml per hour	8 ml	15 minutes
F	zero	10 ml	15 minutes

There is evidence that lower background infusion rates and larger PCEA boluses provide better patient satisfaction, with improved analgesia and mobility. For most women, programme E is appropriate.

The pre-programmed pump settings may be altered by the Anaesthetist eg for women of very short stature or those for whom the standard settings are not proving effective.

Pump settings, prescriptions, infusions and connections must be checked by the Anaesthetist AND Midwife, and should be rechecked by the 2 Midwives at each handover. Any alteration to the pump settings must be prescribed, checked and signed similarly.

Check the keypad lock on the pump is active.

Connect the pump handset and give it to the woman. Supervised by the midwife, she may push the button to give herself a bolus at any time when she feels that pain relief is inadequate. The handset button may also be pushed by the Midwife.

5.10. Prescriptions by the Anaesthetist

Initial dose(s) used to establish analgesia should be fully prescribed and signed as given on the 'once only' section of the Prescription Chart.

The PCEA prescription is written by the Anaesthetist in the 'as required' section of the Prescription Chart, with all pump settings included, and is checked and signed by Anaesthetist and Midwife when the pump is started. All connections in the epidural catheter pathway are checked by Anaesthetist and Midwife, and presence of yellow epidural colouration / stickers checked.

The Anaesthetist should ensure that an iv infusion of Plasmalyte is prescribed, and that oral antacid prophylaxis is prescribed unless contraindicated.

The Anaesthetist should also usually prescribe in the 'once only' section of the Prescription Chart:

- 0.1% Levobupivacaine 5+5 ml epidural PCEA 'Clinician Bolus' if required

This 'clinician bolus' may then be administered via the PCEA pump by a Senior Midwife, as an extra top-up if analgesia is inadequate. For monitoring and actions if this is used see below.

5.11. Monitoring

The Anaesthetist must be immediately available on delivery Suite for at least 15 minutes after the loading dose or any extra top-ups given by the Anaesthetist, to check the effect and manage any complications.

1:1 Midwifery care required throughout, with continuous CTG monitoring

BP check before and every 5 minutes after for the first 20 minutes after any loading dose/top-up given by Anaesthetist/Clinician Bolus given by Senior Midwife, recorded on epidural chart

During PCEA use, every 30 minutes, record

Number of PCEA boluses

Volume of epidural infused

Maternal pulse and blood pressure

Foetal heart rate

Analgesia

Motor weakness – if the woman is unable to lift her heel off the bed, inform the Anaesthetist

Level of sensory block (to cold) – if the level of sensory block is up to T6 (xiphisternum or underside of breasts), stop the PCEA pump and contact the Anaesthetist

5.12. General Midwifery Management of Woman with Epidural

- Maintain patent iv access
- 1:1 Midwifery care
- Fluid balance chart
- Oral intake: antacid prophylaxis as current protocol, and if labour is complicated or there is significant risk of Caesarean Section restrict oral intake to fluids only
- Bladder Care: epidural may interfere with bladder sensation. Encourage woman to empty her bladder. May require in-out or indwelling catheter if unable to pass urine spontaneously within 2 hours of epidural insertion
- Mobility and Positioning: woman must remain in bed, with compression stockings. Avoid supine hypotension/aortocaval compression. Maternal position must be changed regularly to prevent pressure area damage. Using 0.1% Levobupivacaine, any leg weakness should be mild, and the woman should be able to move herself around in the bed with minimal assistance, but she must reposition at least hourly. If she is unable to move herself, check block height and contact Anaesthetist.

- Continue epidural infusion until completion of the Third Stage of Labour and any necessary perineal repair. Consider indwelling catheter insertion after delivery if catheter has been needed in labour, and for obstetric indications.

- Regular repositioning must continue after delivery until the epidural analgesia has fully resolved and the woman can mobilise normally. If the woman cannot lift each leg off the bed 2 hours after stopping the epidural infusion, contact the Duty Obstetric Anaesthetist immediately.

5.13. Inadequate Pain Relief

- Try to establish the possible cause: check skin insertion site, all connections, map the distribution of pain and the extent of the block by testing with ice.

- Note that 5-8 ml per hour background infusion will not maintain analgesia in most labouring women – they should expect to need to use the PCEA button

- For breakthrough pain in an established effective epidural, encourage PCEA use to give a bolus of the infusion mixture and repeat as necessary.

- If this is not effective the Midwife should discuss with the Senior Midwife, and consider contacting the on call Anaesthetist, and/or giving the prescribed 'Clinician Bolus' depending on the circumstances.

- If analgesia is still inadequate, the Anaesthetist should be informed.

- For any topups delivered by syringe, sterility of all connections must be maintained, and the epidural catheter should be aspirated carefully (for at least 30 seconds) to check for cerebrospinal fluid or blood before injection. Any syringe containing local anaesthetic for epidural injection should be labelled with a yellow 'epidural use only' sticker.

5.14. Procedures in Theatre

- An effective labour epidural may be topped up by the Anaesthetist for most Obstetric procedures in theatre, including for Caesarean Section (see section 5.2)

- Contact the Anaesthetist as early as possible

5.15. Complications of Epidural Analgesia

- For any woman about whom the Midwife is concerned, inform the Senior Midwife and consider contacting the Obstetric Anaesthetist and/or Obstetric Emergency Team and stopping the PCEA pump.
- Use ABC resuscitation approach, and nurse the woman in a position to avoid aortocaval compression.
- See Appendix 3 for further details.

5.16. Epidurals and Coagulopathy

- A platelet count and coagulation screen may be requested by the Anaesthetist before siting an epidural. The Anaesthetist must take into account the clinical situation and potential changes/rate of change, and results should be recent, depending on the clinical picture. If in doubt, contact the on call Consultant Anaesthetist. If a 'routine' Full Blood Count has been taken, the anaesthetist will use their clinical discretion, and may wait for the result before siting an epidural, including in an otherwise fit and well woman in normal labour.
- Platelet count and antiplatelet drugs:

Platelet count	epidural
<70 $x10^9$ per litre	contraindicated
70-100 $x10^9$ per litre	Relatively contraindicated, seek advice from consultant anaesthetist
> 100 $x10^9$ per litre	Generally thought to be safe, but consider trend and platelet function

Clopidogrel should be stopped 7 days before an epidural

Aspirin 75-150 mg daily is not a contraindication to epidural

- Women on heparin or warfarin (including Low Molecular Weight Heparin LMWH)
  - Most women on antenatal anticoagulation will have been seen in the Anaesthetist Obstetric Clinic and will have a plan in the notes related to options for analgesia and anaesthesia. However consideration should always be given to the CURRENT circumstances, and this plan may need to be reconsidered. If in doubt, discuss with the Consultant Anaesthetist on call.

Full anticoagulation: with heparin, LMWH, or warfarin: epidural contraindicated until reversed and INR<1.4 and APTT ratio <1.3. Discuss with Consultant Anaesthetist.

- For all doses/medications, if epidural insertion is difficult or there is a 'bloody tap', consider delaying the next dose of anticoagulant medication for a longer period after epidural insertion or removal. Consider alternative DVT prophylaxis etc to cover this time interval eg compression stockings, pneumatic boots/leggings if appropriate.

- The likelihood of epidural haematoma is relatively increased in all women with coagulation abnormalities: all should be observed closely for symptoms and signs of this complication, and the Obstetric Anaesthetist must be contacted without delay if a block is unexpectedly dense, does not resolve in a normal way, or there are concerns. See below for monitoring resolution of block.

- **Prophylactic Unfractionated Heparin**: wait 4 hours after last dose before epidural insertion or removal and wait at least 2 hours after insertion or removal before next dose

- **Prophylactic LMWH** (eg 20-40 mg enoxaparin once daily): wait 12 hours after last dose before epidural insertion or removal and allow at least 4 hours after insertion or removal before the next dose

- **Prophylactic LMWH, adjusted for raised BMI women** (eg enoxaparin 60 mg once daily): There is little evidence regarding safety and timings for procedures following these increased, weight adjusted 'prophylactic' doses. Consider risks and benefits for each individual woman and if necessary discuss with the on call Consultant Anaesthetist. An epidural should not be sited less than 12 hours after the last dose, and no further dose should be given for at least 4 hours after insertion or removal of epidural catheter.

- **High dose ('treatment dose') LMWH**: An epidural should not be sited less than 24 hours after the last dose, and no further dose should be given for at least 4 hours after insertion or removal of epidural catheter. Check coagulation screen and platelet count and discuss with on call Consultant Anaesthetist before siting an epidural.

5.17. Accidental disconnection of epidural catheters

- Disconnections may be reduced by careful checking at initial siting of epidural. Consider using sterile dressing to reinforce the connections points.
- Connections from epidural bag via tubing to filter, connector and epidural catheter should be checked by the midwife half hourly, along with other epidural checks
- If a disconnection is identified, risks include failure of pain relief, and infection
- Midwife should stop the PCEA pump, wrap the two disconnected ends in sterile swabs, assess the effective block and contact the Obstetric Anaesthetist.
- The Anaesthetist will review, and make a decision with the patient as to whether to continue using the epidural catheter after cleaning/cutting/changing any parts, or whether to remove the epidural and resite or offer alternative pain relief. An epidural top up may be needed to re-establish analgesia.
- Where there has been disconnection between the epidural catheter itself and the catheter clamp, risks of infection should be considered and discussed with the patient, but in most cases the epidural catheter may be cut about 5 cm proximal to the tip, and joined to a new replacement clamp. Chlorhexidine should NOT be used to wipe the epidural catheter. Saline may be used if necessary.
- The anaesthetist must make a written record in the notes AND record the disconnection and subsequent actions via a Datix Incident Report.

5.18. Responsibilities

- Anaesthetist: ensure the woman is suitable for epidural, obtain and document informed consent, site the epidural and initiate analgesia, call for senior assistance if unsuccessful after 2 attempts or if complications arise, check block height after the first dose, set up PCEA pump and check settings, connections and bag of local anaesthetic with midwife, ensure yellow ‘epidural only’ sticker is placed on the epidural catheter tubing, prescribe all medications and fully complete the anaesthetic section of epidural monitoring chart, complete the anaesthetic audit chart and place it in the follow up folder in the anaesthetic room

- Midwife: monitor the woman & epidural pump and document, give 'clinician bolus' as prescribed with the Senior Midwife, recognise complications, call for assistance and initiate resuscitation if required, keep the Obstetric Anaesthetist informed of any complications or problems including inadequate pain relief, remove the epidural catheter at the appropriate time and record its complete removal in the notes, make an assessment of the epidural, record mode of delivery and file monitoring form in medical notes (with copy in epidural file in Delivery Suite Office), monitor post epidural for expected resolution of block and raise concern if appropriate.

6. Operative Procedures

Instrumental Deliveries and Suturing of Perineum in Delivery Room

6.1.1. Because instrumental birth is an operative procedure, advise the woman to have effective tested anaesthesia. If a woman declines anaesthesia, offer a pudendal block combined with local anaesthetic to the perineum. See Appendix 1 for process.

6.1.2. Ensure appropriate anaesthesia is in place for mid-cavity rotational deliveries. This will usually be a regional block (spinal or epidural anaesthesia). A pudendal block may be appropriate, particularly in the context of urgent delivery.

6.1.3. If there is concern about foetal compromise, offer either tested effective anaesthesia or, if time does not allow this, a pudendal block combined with local anaesthetic to the perineum.

6.1.4. Note that for those women with epidural analgesia for labour, local anaesthetic doses used for labour may NOT be adequate for instrumental delivery, and a top up may be required. It may be more appropriate to move the woman to theatre for full regional anaesthesia.

6.1.5. Analgesia after instrumental delivery: regular paracetamol and NSAID have been shown to be beneficial after caesarean section and for perineal pain. They should be considered after operative vaginal delivery.

Caesarean Deliveries

The preferable technique for most women is regional anaesthesia

6.2.1. Caesarean with Regional Anaesthesia

6.2.1.1. Epidural Anaesthesia

An effective labour epidural may be toped up by the Anaesthetist. Contact the Obstetric Anaesthetist as early as possible.

Give oral antacid prophylaxis if not already given, and sodium citrate 0.3M 30 ml orally

The anaesthetist will assess the woman and decide whether to top up the existing epidural, perform a spinal, or use general anaesthesia. The Obstetrician should be consulted to establish degree of urgency so that the anaesthetist can then use the most appropriate technique. If an epidural has not been effective for labour, then alternative anaesthesia is preferable: use spinal or general anaesthesia as appropriate.

For epidural top up anaesthesia, the woman should be informed as a minimum about nausea/vomiting, sensations of tugging and pulling, hypotension, and possible need for conversion to general anaesthesia, and specifics of the consent must be recorded by the Anaesthetist.

Suitable epidural topups for procedures in theatre include:

0.5% levobupivacaine 4+16 ml

2% Lidocaine with 1 in 200 000 adrenaline 4+16 ml

2% lidocaine with 1 in 200 000 adrenaline and 1-2ml 8.4% sodium bicarbonate 4+16 ml

Opioids eg diamorphine 3 mg or fentanyl 1 mcg per kg should usually be added to the above local anaesthetic solutions.

All epidural topups for procedures in theatre should be administered in theatre with essential monitoring established.

All syringes intended for epidural topup should be identified with yellow coloured stickers.

Sterility of all connections must be maintained.

In all cases, an initial ‘test’ dose should be used (eg 3-4 ml of the chosen solution) to identify subarachnoid catheter before giving the full dose. It is likely that close to 20 ml total local anaesthetic will be needed for effective anaesthesia.

The anaesthetist must check the epidural anaesthesia is adequate before allowing surgery to start, and make a record of this check. Checks should include assessment and record of block height, motor block, sacral area block.

If the block is deemed inadequate for surgery, surgery must not start. The anaesthetist and woman will decide whether to give further epidural topup, site spinal or give general anaesthetic, and this decision will be influenced by degree of urgency of delivery.

If the women is in pain during the procedure at any stage, surgery must be stopped while the anaesthetist assesses the situation. The women should be offered a general anaesthetic, and the anaesthetists must record this, even if the woman declines. Depending on circumstances, analgesia may be offered, but if the pain continues, general anaesthesia should e recommended to the woman, and if she declines this must be recorded. The on call Consultant Anaesthetist should be informed of any case where epidural anaesthesia is inadequate and general anaesthesia is refused by the woman.

6.2.1.2. Spinal Anaesthesia

For women without effective epidural analgesia, and for women wh ar not in labour, in most cases spinal anaesthesia is the safest and preferred technique for Caesarean Delivery.

The anaesthetist will assess the woman and explain the procedure. The discussion of risks and benefits must be recorded in the notes, and should include as a minimum: severe headache, hypotension with nausea and vomiting, sensation of tugging/pulling/pressure nerve damage, failure of technique and option for general anaesthesia. In emergency cases the consent process may need to be adjusted to the circumstances and a record made.

Spinal anaesthesia must be performed with standard monitoring in place, avoiding aortocaval compression at all times.

In most cases, the anaesthetist will use full aseptic technique, with gown, mask, hat and gloves. In very urgent cases, a gown may not be used.

Skin preparation is with Chlorhexidine Gluconate 0.5%, and allowed to dry.

Local anaesthetic to skin and subcutaneous tissues may occasionally be omitted in very urgent cases.

For most women, 2.2-2.6ml 0.5% 'heavy bupivacaine' with opioid diamorphine 0.3 mg is appropriate. Diamorphine should be reconstituted with 0.9% saline. Fentanyl 15-25 mcg may be used as alternative to diamorphine.

Opioids contribute to effective anaesthesia during the procedure, allowing relatively low doses of local anaesthetic and hence less hypotension. Diamorphine will also provide up to about 12 hours of analgesia post-operatively at this dose. Fentanyl will provide little post-operative pain relief. The woman must be monitored post-operatively for delayed respiratory depression which may occur with fentanyl or diamorphine.

6.2.2. Caesarean under General Anaesthesia

May be required if there is too little time to establish a regional block, if the woman declines regional anaesthesia, or for obstetric or medical indications. Consent should include explanation of pre-oxygenation and cricoid pressure. Most cases are emergency procedures. The anaesthetist will assess the woman and use the most appropriate anaesthetic under the circumstances, and record details of the consent process.

A paediatrician will be present at deliveries under general anaesthesia

Antacid prophylaxis will be administered orally as soon as possible after the decision for caesarean is made, if not had it already. All women should receive 30 ml 0.3M Sodium Citrate shortly before anaesthesia.

Women are anaesthetised in theatre with routine monitoring established. Aortocaval compression must be minimised. Rapid Sequence Induction with Pre-oxygenation and cricoid pressure is used. Consider using videolaryngoscopy.

Maintenance of anaesthesia is usually with inhalational agent and N2O with oxygen. Opioids are often withheld until after delivery: if they are used before delivery, inform the Paediatrician.

Complications are more common at emergence than at induction, with high risk of regurgitation and aspiration and other airway complications. Women should be extubated in theatre, awake, with an experienced anaesthetist present.

6.2.3. Pain Relief after Caesarean Delivery

Most women who have had regional anaesthesia will have already received neuraxial opioids and PR diclofenac 100mg (which must be recorded as a 'once only' prescription on the woman's prescription chart). Neuraxial diamorphine provides up to about 12 hours of pain relief whether given spinally (0.3 mg) or epidurally (3 mg). If larger doses are used, duration of analgesia is increased but with increased opioid side effects especially pruritis, and risk of respiratory depression.

In the absence of contraindications, the following should be prescribed by the anaesthetist before leaving theatre:

1. Paracetamol 1g regularly 6 hrly, oral/iv, and any doses already given clearly marked in the Prescription chart
2. Ibuprofen 600 mg regularly 8hrly, with the first dose to be administered at a suitable time after the PR diclofenac given in theatre
3. Oramorph (oral morphine) 10-20 mg 2 hourly as required

Non Steroidal Anti-inflammatory Drugs (NSAIDs) provide effective pain relief, including in breast feeding women. Prescribe with caution if haemorrhage has been significant or there is ongoing risk of haemorrhage, and in Pre-eclampsia.

Codeine phosphate 30-60 mg is not recommended for breast feeding women. It has been associated with significant respiratory depression in neonates.

After general anaesthesia, it may be more appropriate to use morphine as iv PCA than oral morphine, as these patients will not have received neuraxial opioids. See *Acute Pain Guidelines*. Women should be converted to oral analgesia as soon as possible.

Local anaesthetic techniques such as Transversus Abdominis Plane (TAP) block can provide additional post operative analgesia, and are particularly useful for patients who have not had neuraxial opioids. TAP catheters/wound catheters/rectus sheath catheters can be used for longer benefit compared with single shot techniques but these are not routine techniques in the Maternity Unit: discuss with the Midwifery team and Consultant Anaesthetist in advance, involve the Pain Team, and ensure that written instructions are available for Midwifery staff caring for any woman with nerve or wound catheters in situ.

Other Operative Procedures

6.3.1. Manual Removal of Placenta

Regional or General Anaesthesia may be offered.

For regional anaesthesia, the anaesthetist must specifically discuss/record in the consent process the marked sensation of tugging, pulling and pressure the woman may experience, and must ensure adequate density and extent of anaesthesia and record this before allowing the procedure to start. General anaesthesia should be offered during the procedure if the woman requires it.

For epidural topup, local anaesthetic as listed above under Caesarean Section is appropriate. Neuraxial opioids may be added, but postoperative pain is not marked.

For spinal anaesthesia, doses as for Caesarean are recommended similarly.

General Anaesthesia carries greater risk of serious complications for the woman, but may be preferred by some women. Use Rapid Sequence Induction, with Cricoid Pressure as for Caesarean.

6.3.2. Instrumental Delivery in theatre, Suturing of Perineum in theatre

Spinal anaesthesia or epidural topup may be used, using local anaesthetics and opioids as listed above under Caesarean Delivery. Postoperative pain is variable: analgesia must be prescribed.

Occasionally general anaesthesia may be used: use Rapid Sequence Induction, with Cricoid Pressure as for Caesarean.

6.3.3. Post Partum Haemorrhage Anaesthesia

Depending or clinical circumstances regional or general anaesthetic (with Rapid Sequence Induction, with Cricoid Pressure as for Caesarean) may be appropriate.

Consider informing on call Consultant Anaesthetist if there has been significant haemorrhage, or if it is anticipated.

See *Obstetric Haemorrhage Guideline* and *Massive Blood Loss Management Policy (Adults)*

Recovery after Anaesthesia in the Maternity Unit

All women who have had anaesthesia in the operating theatre will recover under the direct care of a trained Recovery Nurse/Practitioner. Recovery may be in theatre, Recovery area or Delivery Suite, depending on circumstances. At the end of the recovery period, the Recovery Practitioner will hand over care to the Midwifery Team.

Babies will wherever possible stay with their mother, and Recovery staff will facilitate this. The baby remains throughout under the direct care of the Midwife, and she must remain within immediate calling distance at all times.

See *Post Anaesthetics Care Unit (PACU) Discharge Criteria Guidelines* for further information.

7. Monitoring Compliance and Effectiveness of the Guideline

Process for Implementation and Monitoring Compliance and Effectiveness

Will be the responsibility of the Lead Clinician for Obstetric Anaesthesia, Head of Midwifery and Lead Obstetrician for the Delivery Suite.

An up to date copy of the guideline is available to all staff on the Trust intranet. It will be reviewed tri-ennially by the Maternity Services Governance Group.

Reporting of non-compliance and review of effectiveness will be identified through the risk process within the maternity unit (via Maternity Services Governance Group), and led by appointed midwifery, obstetric and anaesthetic risk leads.

All versions of this guideline will be archived in electronic format by the author within the Maternity Team Policy Archive. Any revisions will be recorded on the Document Control Report. To obtain a copy of the archived guidelines, contact should be made with the Maternity Risk Manager.

- Any Datix incident reports will be reviewed by Governance leads for Obstetrics, Midwifery & Anaesthetics as appropriate
- Issues and learning points will be discussed between and within the relevant teams through routine departmental Governance meetings and any actions implemented and confirmed as complete

➢ Obstetric Audit and Case Review meetings

- Anaesthetic Morbidity and Mortality meetings

8. References

- Regional Anaesthesia and Patients with Abnormalities of Coagulation https://anaesthetists.org/Portals/0/PDFs/Guidelines%20PDFs/Guideline_regional_anaesthesia_patients_abnormalities_coagulation_2013_final.pdf?ver=2018-07-11-163756-520&ver=2018-07-11-163756-520

- Non-regional analgesia for labour: remifentanil in obstetrics https://www.bjaed.org/article/S2058-5349(19)30114-3/fulltext

- https://pubmed.ncbi.nlm.nih.gov/24418650/ Influence of protective measures after epidural catheter disconnection on catheter lumen colonization: an in vitro study

- Management of disconnection between epidural filters and catheters: a dilemma for the anaesthetist https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/full/10.1046/j.1365-2044.2000.055002192.x

- https://www.labourpains.com/Epidural-Information-Card

- Guidelines for the Provision of Anaesthesia Services Chapter 9: Guidelines for the Provision of Anaesthesia Services for an Obstetric Population https://rcoa.ac.uk/gpas/chapter-9

9. Associated Documentation

- Acute Pain Guidelines
- Obstetric Haemorrhage Guideline
- Massive Blood Loss Management Policy (Adults)
- Post Anaesthetics Care Unit (PACU) Discharge Criteria Guidelines
- Intrapartum care: care of healthy women and their babies during childbirth Including Fetal Monitoring in Labour
- Shoulder Dystocia Guidelines

10. Appendices

10.1. Pudendal nerve block

10.2. Remifentanil Patient Controlled Analgesia PCA

10.3. Complications of epidural analgesia

11. PUBLICATION DETAILS

Author of Clinical Guideline	Consultant Anaesthetist
Directorate/Department responsible for Clinical Guideline	Clinical Support and Specialist Services / Maternity Services
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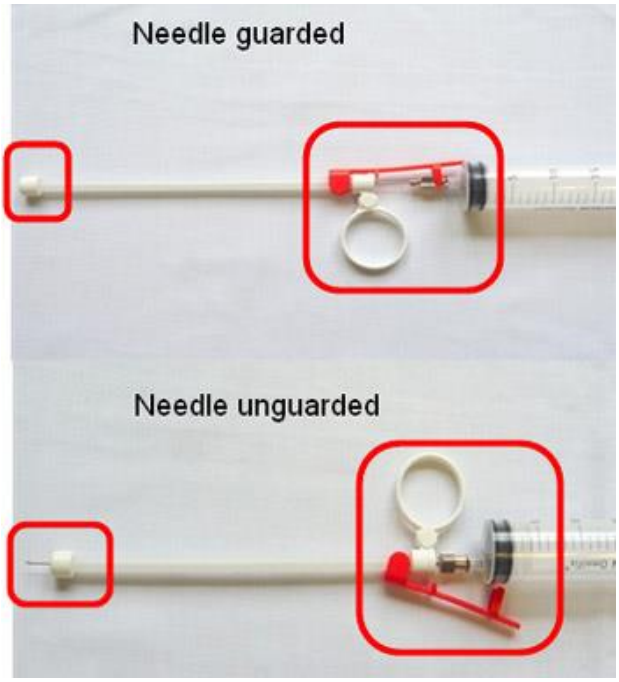
Appendix 1 Pudendal Nerve Block

Draw up 20 ml 1% lidocaine into a syringe.

It is preferable to use a special needle where the point is guarded. This limits the depth of penetration to 1 cm (where the pudendal nerve is located) and helps avoid needlestick injuries.



When a left-sided block is being performed, the ischial spine is palpated with the index or middle finger of the left hand and vice-versa for a right-sided block. The syringe is held with the other hand and the guarded needle is guided between the index and middle fingers.

There are two accepted methods for performing a para-vaginal pudendal block.

Position your index finger on the ischial spine and run the needle guard in between (see image below) your index and middle fingers. Place the end of the needle guard 1 cm posterior and medial to the ischial spine. The guard can then be unhooked and the needle advanced a centimetre into the tissue.

Position your middle finger on the ischial spine and run the needle guard in between (see image below) your index and middle fingers. Place the end of the needle guard 1 cm anterior and medial to the ischial spine. The guard can then be unhooked and the needle advanced a centimetre into the tissue.

Large pudendal vessels lie in close proximetry to the nerve. It is therefore very important to aspirate the syringe (to check that the tip of the needle is not in a vessel) prior to injecting local anaesthetic. After aspiration, inject 7 ml of lidocaine 1%.

The block should then be repeated on the other side. The remaining 6 ml of lidocaine should used to infiltrate the perineum.

Reference: https://stratog.rcog.org.uk/tutorial/easi-resource/performing-a-pudendal-block-5134

Appendix 2 Remifentanil PCA patient information leaflet

What is remifentanil?

Remifentanil PCA is a type of pain relief sometimes used for women during labour. Remifentanil is classed as an opioid, (a morphine-like drug), and is able to reduce the severity of pain during labour. Although it cannot provide complete pain relief many women find it helpful. Remifentanil is a very short-acting pain relieving drug; its effect comes on very rapidly and wears off very quickly and so it must be given through an electronic pump attached to a drip in your arm. Remifentanil has been used in labouring women in the UK for special indications for more than 15 years with good results and favourable feedback.

Who can use remifentanil?

In our hospital, remifentanil PCA may be used in your labour if you are unable to have an epidural or if epidural has not been effective for you. If you have an allergy to morphine, pethidine or other related drugs we may advise you not to use remifentanil. Remifentanil is not suitable if you have been given pethidine or diamorphine within the last four hours. You may still use gas and air (entonox) at any time. Remifentanil may be useful if you have been advised not to have an epidural. If you try remifentanil and then decide to change to another method of pain relief all other options are still open to you.

How is it given?

To use remifentanil you will need to have a cannula (commonly known as a "drip") placed in a vein, usually on the back of your hand or arm. The drip is connected to an electronic pump, which delivers a small dose of the drug when you press the hand-held button. The pain relieving effect is felt usually within 20 – 30 seconds, and wears off again within a few minutes. As part of our routine observations with remifentanil your midwife will measure your oxygen level using a sensor (like a clothes peg) on your finger, as well as your level of pain relief and drowsiness at regular intervals. Otherwise all observations and treatment are similar for any other woman on the Delivery Suite.

What are the potential benefits of remifentanil?

You are in control and you only get the drug when you need it and not in between contractions. This means that you may be more alert and mobile throughout your labour. There is a safety feature built into the pump so that you can only get a safe amount of the drug. You can use the pump at any time right up to your delivery (and for stitches if needed) and the effects will still wear off very quickly when you stop.

Are there any unwanted effects of remifentanil?

Remifentanil can make you feel dizzy and sleepy between contractions. This will wear off very quickly when you stop using the pump. Remifentanil can cause the oxygen levels in your blood to fall. If this occurs you may need to have additional oxygen given. Rarely, the effect on your breathing may be more troublesome, possibly dangerous, and you may need to stop using the PCA and/or receive a medicine to counteract these effects. Occasionally itching and nausea can occur. These unwanted effects are similar to those found with pethidine and diamorphine.

How will it affect the baby?

Remifentanil has been shown to be safe for babies. There are no other effects that don't already occur with pethidine and diamorphine. Any remifentanil effects wear off as quickly as in an adult.

When can I ask for remifentanil?

A remifentanil PCA can be started at any point in your labour. Your midwife will call the anaesthetist to discuss it with you, and set up the pump for you. This may take a few minutes, but you will be able to use it immediately once it has been connected to the drip.

Discussion before using remifentanil

You should read this leaflet carefully. The anaesthetist will discuss it with you, and answer any questions you have. He/she will make sure you understand how to use it, and what risks there are. If you decide to use it, the anaesthetist will set it up. He/she will help you to start with so you can use it effectively and safely.

Any questions?

We are very happy to discuss any questions you have about remifentanil. You can discuss it with your midwife, or with one of our anaesthetic doctors.

Leaflet author:
Dr C Don
Lead Consultant, Obstetric Anaesthesia
Sept 2017 version

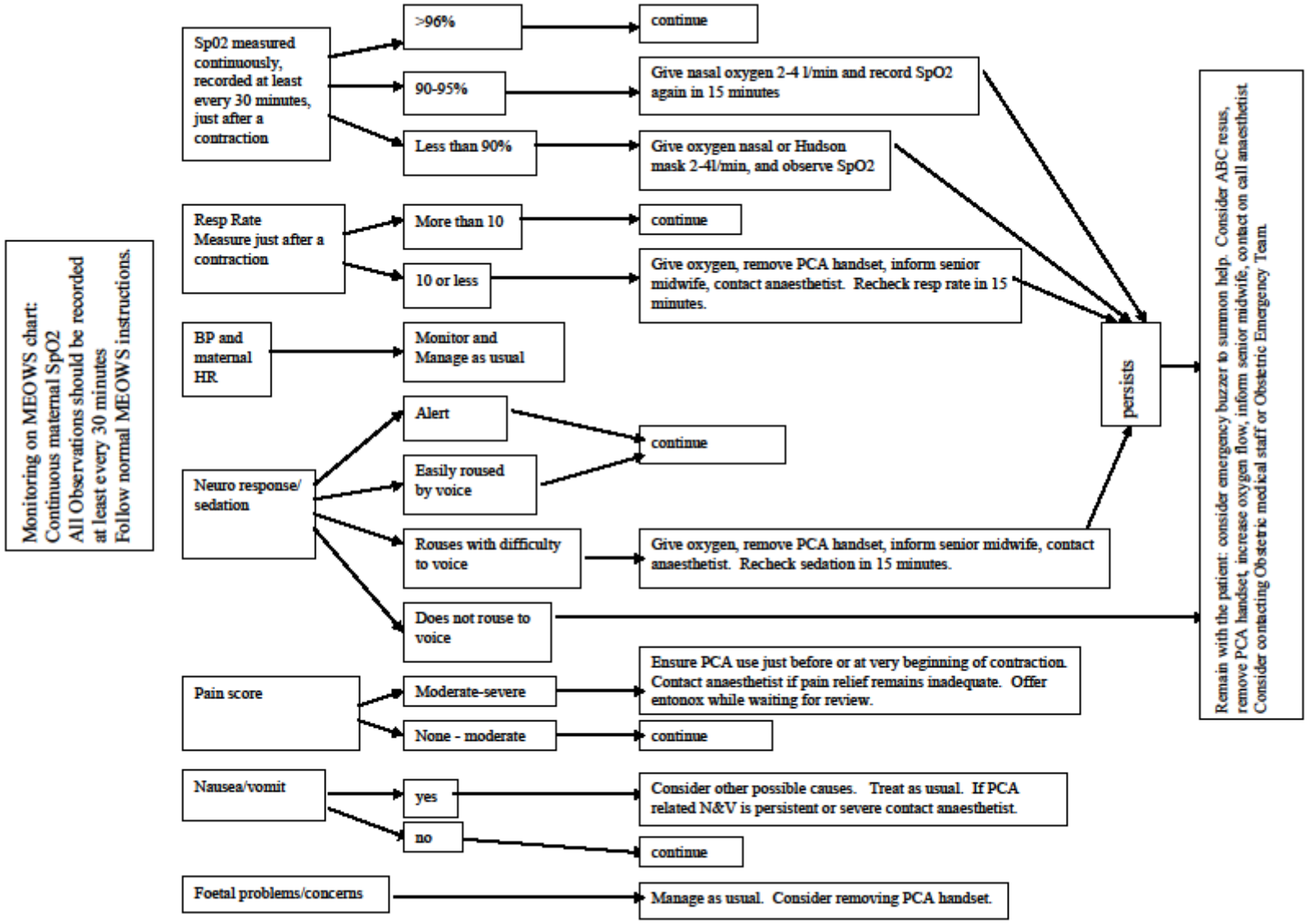
Royal Devon University Healthcare NHS Foundation Trust

Remifentanil PCA

Intravenous Patient Controlled Analgesia in labouring women

Information sheet for women

Remi PCA management flow chart


Monitoring on MEOWS chart:
Continuous maternal SpO2
All Observations should be recorded at least every 30 minutes
Follow normal MEOWS instructions.
Sp02 measured continuously, recorded at least every 30 minutes, just after a contraction
>96%
continue
90-95%
Give nasal oxygen 2-4 l/min and record SpO2 again in 15 minutes
Less than 90%
Give oxygen nasal or Hudson mask 2-4l/min, and observe SpO2
Resp Rate Measure just after a contraction
More than 10
continue
10 or less
Give oxygen, remove PCA handset, inform senior midwife, contact anaesthetist. Recheck resp rate in 15 minutes.
BP and maternal HR
Monitor and Manage as usual
Neuro response/ sedation
Alert
Easily roused by voice
continue
Rouses with difficulty to voice
Give oxygen, remove PCA handset, inform senior midwife, contact anaesthetist. Recheck sedation in 15 minutes.
Does not rouse to voice
persists
Remain with the patient: consider emergency buzzer to summon help. Consider ABC resus, remove PCA handset, increase oxygen flow, inform senior midwife, contact on call anaesthetist. Consider contacting Obstetric medical staff or Obstetric Emergency Team.
Pain score
Moderate-severe
Ensure PCA use just before or at very beginning of contraction. Contact anaesthetist if pain relief remains inadequate. Offer entonox while waiting for review.
None - moderate
continue
Nausea/vomit
yes
Consider other possible causes. Treat as usual. If PCA related N&V is persistent or severe contact anaesthetist.
no
continue
Foetal problems/concerns
Manage as usual. Consider removing PCA handset.

Remifentanil IV PCA
Quick reference guide

For detailed guidance see 'Bob': "Analgesia and Anaesthesia in Maternity Guideline".

1:1 midwifery care at all times.
Mother must be 37+ weeks (i.e. must have completed 36 weeks pregnancy).
Consultant Anaesthetist on call must be informed.
Remifentanil may not be started if parenteral opioids have been given in the last 2 hours
Never leave the patient alone

Dedicated iv cannula for iv PCA e.g. 20G (pink) – NO other drugs or fluids to be administered through this cannula.
IV fluids, syntocinon etc. are administered as usual through a separate (usually 16G) cannula, which MAY be in the same arm.
Means to give facial oxygen must be in the delivery room: advanced resus equipment available on Delivery Suite resus trolley/emergency airway box as usual.

Remifentanil is reconstituted by the anaesthetist, checked with the midwife.
2mg Remifentanil in 50ml 0.9% Saline producing final concentration 40 micrograms per ml.

Bolus dose of 20, 30 or 40 micrograms remifentanil may be prescribed by the anaesthetist.
The 'lockout' time is fixed in this pump at 2 minutes.

The PCA pump must be locked via keypad (by anaesthetist) AND syringe clamp locked with key (held by anaesthetist and midwife).

Only the patient may push the PCA button. It must not be used by birth partner or midwife.

All patient observations are recorded on the MEOWS chart, half hourly or more frequently, following the flow chart overleaf.
Observations for the PCA pump itself are recorded on the reverse side of the MEOWS chart, at least half hourly.

The midwifery team may make up a new syringe of remifentanil if the first syringe runs out. This will be of the same concentration already prescribed by the anaesthetist, and the syringe and all pump settings are checked with the Senior Midwife. The syringe clamp on the pump must be locked in place after changing the syringe, and the keypad locked activated.

The dose of remifentanil may only be changed by the anaesthetist.

On call anaesthetist should be contacted according to the flow chart overleaf, or at any time if the midwife is concerned.

On call anaesthetist:	Epidural anaesthetist	bleep 234	Obstetrician:	Duty Obstetrician	bleep 013
	Obstetric Anaesthetist	bleep 822		Ward doctor, Obstetrics	bleep 299
	ICU/anaes doctor	bleep 504			

Remi PCA Audit sheet

Remifentanil PCA audit sheet:

MIDWIFE Please complete after delivery and place sheet in anaesthetic audit box in Delivery Suite office

Patient label

Date	
Last diamorph (time) if used	
Time remi started	
Time remi stopped	
Time of delivery	

Reason for PCA remifentanil		
Maternal choice		
Epidural failed		
Epidural contraindicated		
	Spina bifida	
	Spinal surgery	
	Sepsis	
	Thrombocytopaenia	
	Coagulopathy	
	Other (please give details)	

Apgar 1 min	
Apgar 5 min	

Mode of delivery	
SVD	
ventouse	
Forceps in delivery room	
Forceps in theatre	
C section	

Complications	
Itch	
Desaturation	
Lowest saturation (SpO2)	
Difficult to rouse or unrousable at any time	
Oxygen given	
Conversion to epidural	
Nausea	

Mode of analgesia for delivery	
None	
Remifentanil	
Entonox	
Spinal	
Epidural	
CSE	
GA	

Any comments:

Midwife's assessment of pain relief	
Labour	
Delivery	
Time of epidural if applicable	

Maternal assessment	
Very Satisfied	
Satisfied	
Not satisfied	

Appendix 3 Complications of Epidural Analgesia

For any woman about whom the midwife is concerned, inform the Senior Midwife and consider contacting the Obstetric Anaesthetist and/or Obstetric Emergency Team and stopping the PCEA pump.

Use ABC resuscitation approach, and nurse the woman in a position to avoid aortocaval compression.

All significant complications must be reported via the Trust Datix Incident Report system

Hypotension

Defined as a fall in systolic blood pressure of 20% from baseline, or below 100 mm Hg, or decrease in blood pressure which is symptomatic. Nausea is a common symptom of hypotension.

Management:

Nurse in full left lateral position to avoid aortocaval compression

ABC approach

Give oxygen by facemask

Give rapid fluid bolus of up to 500 ml crystalloid

Check the block height and consider stopping the PCEA pump

Consider other causes of hypotension e.g. haemorrhage, sepsis, anaphylaxis.

If BP remains low or hypotension thought to be due to epidural, contact the Anaesthetist.

The anaesthetist may give vasopressors eg ephedrine or phenylephrine

Subdural block

A rare condition where the epidural catheter is misplaced, lying between the dura and arachnoid mater.

Characteristics:

Unexpectedly high level of sensory block, sometimes to cervical dermatomes, occurring over 20-30 minutes

Patchy sensory block often with missed segments and persistent pain

Nasal stuffiness and Horner's Syndrome may develop.

The catheter may migrate through arachnoid and into cerebrospinal fluid.

Midwives and Anaesthetists should have a high index of suspicion for a block with a bizarre distribution.

Management:

Call for help, and manage using ABC approach as above.

Stop the PCEA pump.

Contact the Anaesthetist.

The epidural may need to be resited.

High Spinal, may progress to Total Spinal

The epidural catheter is misplaced into subarachnoid space. If every top up is given in smaller divided doses, a total spinal is very unlikely to occur.

Characteristics:

Rapidly rising block with sensory and motor block spreading to arms and proceeding to respiratory weakness and arrest, cardiovascular collapse and unconsciousness.

Management:

Call for help as above. Consider calling Cardiac Arrest Team, and possible need for urgent delivery (Theatre Team, Paediatric/Neonatal team).

Stop the PCEA pump/any epidural injection.

Initiate ABC resuscitation with oxygen, maintaining airway and ventilation if necessary.

Avoid aortocaval compression and support circulation using IV fluid and Ephedrine.

Resuscitation of the woman takes priority, but the foetus must also be monitored and may need delivery.

Intravenous Injection of Local Anaesthetic

Occurs when the epidural catheter is misplaced into a vein.

Prevented by careful technique, aspiration before injection, use of smaller divided doses, but this is not completely reliable.

Characteristics:

Circum-oral numbness, dizziness, tinnitus, confusion, convulsions, cardiovascular collapse, respiratory arrest.

Management:

Call for help as above. Consider calling Cardiac Arrest Team, and possible need for delivery (Theatre Team, Paediatric/Neonatal team).

Stop the PCEA pump.

Initiate ABC resuscitation with oxygen, maintaining airway and ventilation if necessary.

Avoid aortocaval compression and support circulation using IV fluid and Ephedrine.

Resuscitation of the woman takes priority, but the foetus must also be monitored and may need delivery.

In severe local anaesthetic toxicity (decreased conscious level, convulsions, significant cardiac or respiratory compromise or arrest), consider administration of Intralipid. If there is cardiac arrest, manage this as usual but also administer Intralipid. Please see the Local Anaesthetic Toxicity flowchart in each anaesthetic room or follow this link
https://www.aagbi.org/sites/default/files/la_toxicity_2010_0.pdf

In addition, may need anticonvulsant treatment.

Accidental Dural Puncture

At the time of epidural insertion/use:

CSF is detected draining via the epidural needle on insertion or via the epidural catheter, or following top up it is suspected that the catheter is subarachnoid and not epidural.

The anaesthetist should consider options:

1. Remove epidural needle/catheter, inform woman of complication and consequences, offer resite or alternative analgesia, record in notes and audit form, inform consultant anaesthetist on call, ensure suitable follow up. If resiting, consider resite by more experienced colleague. If resited, initial doses and any top up should be given with caution, by anaesthetist only and not by midwifery staff. The PCEA may be used as normal if the response to the initial dose is normal.

2. Site subarachnoid catheter: insert the catheter no more than 3 cm beyond the tip of the Tuohy needle. Fasten in place as usual. Inform the woman of the complication. Discuss with the on call consultant anaesthetist. There are several options for ongoing management by the anaesthetist, to be decided by the Consultant Anaesthetist:

    a. Remove the catheter and offer resite or alternative analgesia as above

    b. Give a suitable dose of local anaesthetic +/- opioid, through the catheter into the subarachnoid space to establish analgesia (eg 0.125% L-bupivacaine 2 ml +/- fentanyl 10-20 mcg, flush with 2 ml 0.9% saline. Then remove the catheter. An epidural may be resited as above, or alternative analgesia may be offered once this has worn off. Monitoring of BP, CTG etc must be used as for epidural initial dose, and anaesthetist must remain for at least 20 minutes after this dose.

    c. Give a suitable dose of local anaesthetic +/- opioid, through the catheter into the subarachnoid space to establish analgesia (eg 0.125% L-bupivacaine 2 ml +/- fentanyl 10-20 mcg, flush with 2 ml 0.9% saline. Monitoring of BP, CTG etc must be used as for epidural initial dose, and anaesthetist must remain for at least 20 minutes after this dose and any subsequent doses. Label the catheter very clearly as 'subarachnoid catheter', ensure that ALL anaesthetists on call and Delivery Suite staff know this, and ensure that if there is any shift change ALL staff know this. Only an experienced anaesthetist may use this subarachnoid catheter, but it can be used for continuing labour analgesia, using appropriate intermittent boluses of local anaesthetic and opioids. It may also be similarly used for operative delivery, suturing etc.

In all cases, the anaesthetist must ensure that explanation is given to the woman, that suitable senior follow-up is arranged, and that there is a full record in the notes and a completed audit chart. The Consultant Anaesthetist must be informed.

Development of Post Dural Puncture Headache:

A postural headache is common after accidental dural puncture: it may be severe enough to prevent the woman from caring for her baby appropriately. It may occur following an unintended dural puncture which was recognised at the time of epidural insertion, or following an apparently normal epidural or following spinal anaesthesia.

All postnatal headaches which are not relieved by simple analgesia should be referred in the first instance to the Obstetric Team for appropriate history and management.

If the headache is postural and follows regional analgesia, the Duty Obstetric Anaesthetist should be contacted to arrange review, management and follow up as needed.

The General On Call Consultant Anaesthetist must be informed about any Post Dural Puncture Headache, and a Datix report must be completed by the Anaesthetist who confirms the diagnosis.

Any suspected Post Dural Puncture Headache detected in the Community should be discussed in the first instance with the Duty Obstetric Anaesthetist. A management plan may include ward/telephone surveillance, and/or readmission (under a named Obstetrician) for specific assessment, investigation and management.

Epidural Haematoma or Abscess

This is a rare but serious complication, and requires urgent diagnosis and management: **delay in definitive treatment will cause permanent neurological damage**.

The diagnosis should be considered in any woman in whom neurological block (motor, sensory or autonomic) does not resolve gradually after stopping an epidural or in any woman who develops new neurological block (motor, sensory or autonomic) following initial return of function after an epidural or spinal.

Other symptoms may include localised back pain and/or fever/sepsis.

Suspected women should be referred immediately to the Obstetric Team for initial urgent assessment, and should also be immediately referred on to the Duty Obstetric Anaesthetist. All suspected cases should be discussed with the General On Call Consultant Anaesthetist.

Appropriate investigation and management must be carried out urgently, and if the diagnosis is confirmed transfer will be discussed with the Neurosurgery centre for on-going care.

Due to the geography of Devon, ANY DELAY IN DIAGNOSIS AND MANAGEMENT is potentially serious, as confirmation requires time to scan etc, and transfer to the Neurosurgical Centre takes significant time.